Course Name: ETHICAL HACKING

Assignment- Week 4

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10 **Total mark: 10 x 1 = 10**

QUESTION 1:

What is the primary purpose of a Hypervisor in a penetration testing lab setup?

a. It encrypts all network traffic between virtual machines.
b. It allows creation and management of multiple virtual machines on a single physical system.
c. It is a hack-proof security layer for the host OS.
d. It automatically patches vulnerabilities on guest machines.
e. None of these.

Correct Answer: b

Detailed Solution: A hypervisor (Virtual Machine Monitor) allows one physical host to create and run multiple virtual machines, sharing the host's resources such as CPU, memory, and storage. It has no role in encryption, security hardening, or automatic patching.
Thus, the correct option is (b).

QUESTION 2:

Which of the following information CANNOT typically be retrieved using a Whois lookup?

a. Domain registration details
b. Name servers
c. Registrant contact information
d. Historical snapshots of the website
e. Domain expiration date

Correct Answer: d

Detailed Solution: A Whois lookup provides registration details, name servers, contact information, and a domain's expiration date. Historical snapshots of a website are retrieved using services like Archive.org's Wayback Machine, not Whois.

Thus, the correct option is (d).

QUESTION 3:

How does host discovery using an ICMP Echo sweep work?

a. An ICMP type-8 request is sent, and an ICMP type-0 reply confirms the host is alive.
b. A TCP SYN packet is sent, and an ACK confirms the host is alive.
c. It broadcasts ARP requests to identify all hosts on the segment.
d. None of these.

Correct Answer: a

Detailed Solution: In ICMP Echo sweep, an ICMP Echo Request (type 8) is sent to the target; receiving an ICMP Echo Reply (type 0) indicates the host is alive. TCP SYN/ACK relates to TCP-based discovery, not ICMP.
Thus, the correct option is (a).

QUESTION 4:

Which NMAP option(s) can be used to perform a TCP-based host discovery sweep?

a. -PE
b. -PS
c. -PA
d. -PU
e. -PM

Correct Answer: b, c

Detailed Solution: -PS (TCP SYN ping) and -PA (TCP ACK ping) are used for TCP-based host discovery sweeps. -PE is used for ICMP Echo sweep, -PU is used for UDP sweep, and -PM is used for ICMP Netmask (non-echo) sweep.

Thus, the correct options are (b) and (c).

QUESTION 5:

During the TCP three-way handshake, what type of message does the client send first to initiate the connection?

a. ACK
b. SYN
c. SYN-ACK
d. RST
e. FIN

Correct Answer: b

Detailed Solution: The client initiates the TCP three-way handshake by sending a SYN packet indicating its intent to establish a connection and the initial sequence number.

Thus, the correct option is (b).

QUESTION 6:

In a TCP SYN (stealth) scan, which packet response(s) indicates that a port is open or closed, respectively?

a. SYN-ACK received indicates the port is open.
b. RST received indicates the port is closed.
c. ACK received indicates the port is open.
d. FIN received indicates the port is open.

Correct Answer: a, b

Detailed Solution: In a TCP SYN scan, receiving a SYN-ACK indicates the port is open, while receiving an RST indicates the port is closed. ACK and FIN packets are not used as indicators in SYN scan responses.

Thus, the correct options are (a) and (b).

QUESTION 7:

Which of the following statement(s) is/are **true** regarding the default networking mode (NAT) in Oracle VirtualBox?

a. It assigns a unique public IP address to each virtual machine.
b. It assigns the same private/virtual IP address to guest operating systems.
c. Virtual machines using this mode can access the internet.
d. It assigns the host system's IP address directly to guest machines.
e. Virtual machines using NAT mode are directly reachable from outside the host by default.

Correct Answer: b, c

Detail Solution: The default NAT networking mode in VirtualBox assigns a virtual/private IP address to guest machines, and it allows internet access from within the guest. It does not assign unique public IPs, does not share the host's own IP address, and guests are not directly reachable from outside by default (NAT hides them behind the host).

Thus, the correct options are (b) and (c).

QUESTION 8:

Which of the following NMAP option(s) can be used for OS and/or service version detection?

a. -sn
b. O
c. sV
d. -Pn
e. A

Correct Answer: b, c, e

Detailed Solution: -O performs OS detection, -sV performs service/version detection, and -A (aggressive scan) combines OS detection, version detection, script scanning, and traceroute. -sn only performs host discovery (skips port scan), and -Pn skips host discovery entirely, neither of which relates to OS/version detection.

Thus, the correct options are (b), (c), and (e).

QUESTION 9:

If ports 80 and 443 are found open on a target system, what can be reasonably inferred?

a. The target is very likely running SSH services.
b. The target is very likely hosting a website (HTTP/HTTPS).
c. The target is running a mail server.
d. The target is running a DNS resolver.
e. None of these.

Correct Answer: b

Detailed Solution: Ports 80 and 443 are the standard ports for HTTP and HTTPS, respectively, so their being open strongly suggests the target is hosting a website. SSH typically uses port 22, mail servers use ports like 25/110/143, and DNS uses port 53 — none of which are indicated here.
Thus, the correct option is (b).

QUESTION 10:

If the --reason option is used along with a SYN scan. Which of the following is/are correct if reason field shows “syn-ack”?

a. The port is open because NMAP received a SYN-ACK response from the target.
b. The port is closed, and no evidence is available.
c. The target sent an RST packet in response.
d. None of these.

Correct Answer: a

Detailed Solution: When --reason shows “syn-ack” for a port marked open, it means NMAP concluded the port is open specifically because it received a SYN-ACK packet from the target in response to its SYN probe, which is the expected behaviour for an open port in a SYN scan.

Thus, the correct option is (a).

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